

Operating System Process Scheduling and Concurrency Control

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1. Fundamentals of Modern Operating Systems

An operating system acts as an intermediary software bridge between physical computer hardware and user software applications, providing process scheduling, virtual memory management, and file system abstractions.

2. Concurrency and Deadlocks

Deadlock occurs in concurrent computing when two or more execution threads are permanently halted because each process waits for a resource held by another process in a circular wait loop.

3. CPU Scheduling Policies

Common CPU scheduling policies include First-Come First-Served, Shortest Job First, Multi-Level Feedback Queues, and Preemptive Round Robin scheduling algorithms designed to optimize throughput and turnaround latency.